Mark schemes

Q1.

(a) (i) $x = 14t$

Accept as components of position vector
Accept x = 15.7 cos 26.6

B1

$y = 7t - 4.9t^2$

Use of equation

M1

3

Substitution

A1

Accept g; accept y = 15.7 sin 26.6t – $\frac{1}{2}gt^2$

(ii) $t = \frac{x}{14}$ $\quad y = 7 \times \frac{x}{14} - 4.9 \times \left(\frac{x}{14}\right)^2$

Use of candidate's coordinates

M1

Substituted; accept unsimplified

A1

$= \frac{x}{2} - 4.9\frac{x^2}{196}$

$y = \frac{20x - x^2}{40}$

AG; convincingly found

A1

3

(iii) $y = 0$ $\quad 20x - x^2 = 0$

Full method for range (t = 1.4286 for range, or t = 0.7143 for greatest height and horizontal distance then doubled, or $R = \frac{V^2 \sin 2\alpha}{g}$)

M1

x =20

AWRT

A1

2

(b) x = 12, y = 2.4

Substitution into appropriate equation(s), (t = 0.857)

M1

For y = 2.4

A1

$\therefore$ under bar

ft provided y > 0

***Alt:** M1 for x = 10 subs and followed by comparison,*
H =2.5 A1, conclusion A1

A1F

3

(c) Air resistance ignored, ball treated as particle…

B1

1

[12]

Q2.

(a) **F** = $\begin{pmatrix} 6 \\ -2.5 \end{pmatrix}$ N

B1 each component

B1 B1

2

(b) |**F**| = $\sqrt{6^2 + (-2.5)^2}$

Must see +

M1

= 6.5 N

ft from vector in (a)

A1F

2

[4]

Q3.

(a) $x = 7t$

B1

$y = 0 + \frac{1}{2}gt^2$

Accept ±

M1

$y = 4.9t^2$

Accept ±

A1

3

(b) $t = \frac{x}{7}$

Attempt at substitution, or use of equation of trajectory with V = 7 & α = 0

M1

$y = 4.9\left(\frac{x}{7}\right)^2$

$y = \frac{x^2}{10}$

CAO

A1

2

(c) $8.1 = \frac{x^2}{10}$

Full method for x, accept ±

M1

$x = 9$m

AWRT 9.0 if two stages used

A1

2

(d) vert: $v^2 = 0 + 2 \times 9.8 \times 8.1$

For M1 Accept ±,
for A1 consistency of signs needed

M1A1

$u = 7$

B1

speed$^2 = (7^2 + 12.6^2)$

M1

speed = 14.4 ms^{-1}

(14.414)

A1F

5

[12]

Q4.

(a) $\mathbf{v} = 6\mathbf{i} + 2t\,\mathbf{j}$

M1: differentiation attempted and vector quantity for **v** *given*

M1A1

2

(b) sp = $\sqrt{(6^2 + 4t^2)}$ ms^{-1}

M1: sum of squares attempted giving scalar expression

A1 : all correct, accept $(2t)^2$

f.t. **v** *with 2 components*

M1A1F

2

(c) $\sqrt{(6^2 + 4t^2)} = 6\sqrt{2}$

o.e.; scalar expression for **v** *in terms of t from (b) used*

M1

$36 + 4t^2 = 36 \times 2$

$t = 3$

f.t. minor slip in (b) provided t is positive solution of quadratic eqn

A1F

2

[6]

Q5.

(a) $\mathbf{r} = \int (4\mathbf{i} - 2t\mathbf{j})\,dt$

attempted

M1

$= 4t\mathbf{i} - t^2\mathbf{j}$ $(+ c)$

A1

$t = 0$, $\mathbf{r} = 8\mathbf{j}$
used

m1

$\mathbf{r} = 4t\mathbf{i} + (8\mathbf{i} - t^2)\mathbf{j}$

A1F
4

(b) $t = 2$, $\mathbf{r} = 8\mathbf{i} + 4\mathbf{j}$

B1F
1

(c) $8 - t^2 = 0$

M1

t = $2\sqrt{2}$ or t = 2.83

A1
2
[7]

Q6.

(a) (i) vertical: $s = ut + \frac{1}{2}at^2$
$0 = 7t - 4.9t^2$
Full method

M1

$0 = t(7 - 4.9t)$
$0 = 7 - 4.9t$

m1

$t = \frac{10}{7}$ sec (1.43)

A1
3

(ii) $OF = 21 \times \frac{10}{7}$

M1

$= 30$ m

A1F

2

(b) vert : $0 = 3.5t - 4.9t^2$

Full method required

M1

$t = \frac{3.5}{4.9}$

m1

$FG = 21 \times \frac{3.5}{4.9}$

M1

$= 15$ m

A1

4

(c) $GH = \frac{1}{2} \times 15 = 7.5$

B1F

$OH = 30 + 15 + 7.5$

M1

$= 52.5$ m

A1F

3

[12]

Q7.

(a) $\mathbf{r} = \int(2t\mathbf{i} + 4\mathbf{j})\,\mathrm{d}t$

Integration attempt for M1

M1

$= t^2\mathbf{i} + 4t\mathbf{j}(+c)$

A1

$t = 0$, $\mathbf{\Gamma} = 2\mathbf{j}$, $2\mathbf{j} = \mathbf{c}$

m1

$\mathbf{\Gamma} = t^2\mathbf{i} + (4t + 2)\mathbf{j}$

Any vector form

A1F

4

(b) $t = 2$, $\mathbf{\Gamma} = 4\mathbf{i} + 10\mathbf{j}$

subs either value of t into **r** *for M1*

M1 A1F

$t = 4$, $\mathbf{\Gamma} = 16\mathrm{i} + 18\mathrm{j}$

ft for vectors with 2 non-zero components

A1F

$= 12\mathbf{i} + 8\mathbf{j}$

subtraction of two vectors

A1F

4

Alternative:

$[t^2\mathbf{i} + 4t\mathbf{j}]_2^4$

M1

$= (16\mathbf{i} + 16\mathbf{j}) - (4\mathbf{i} + 8\mathbf{j})$

A1F A1F

$= 12\mathbf{i} + 8\mathbf{j}$

A1F

subtraction

[8]

Q8.

(a) $x = 7t$, $y = 7t - \dfrac{1}{2} \times 9.8t^2$ $\quad \begin{pmatrix} 7t \\ 7t - 4.9t^2 \end{pmatrix}$

B1 horiz

M1 use of $s = ut + ½at^2$

M1A1 for vert, accept 10 or g

B1M1A1 **3**

(b) $t = \frac{x}{7}$

B1

$y = 7 \times \frac{x}{7} - \frac{1}{2} \times 9.8 \times \frac{x^2}{49}$

M1A1

$y = x - 0.1x^2$

A1 **4**

***Alternative** (full verification method)*
x = 7t used B1
subs into $y = x - 0.1x^2$

M1A1

$y = 7t - 4.9t^2$ as in (a)

A1

***Alternative**: (use equation of path)*

$y = x \tan \alpha - \frac{gx^2}{2v^2 \cos^2 \alpha}$
45° used

B1

subs

M1A1

simplify

A1

(c) $x = 4$, $y = 4 - 0.1 \times 4^2$

or $t = \frac{4}{7}$ (0.571(43))

M1

$y = 2.4$

A1

above wire as 2.4 > 2
ft slip in y provided y positive

A1F

3

(d) $y = 0$

$x - 0.1x^2 = 0$

M1

$x(1 - 0.1x) = 0$

m1 A1

$x = 10$

A1F

4

***Alternative**: (use of coordinate equations)*

$y = 0$, $7t - 4.9t^2 = 0$

M1

$\Rightarrow$ $t = \frac{10}{7}$ *(1.43)*

m1A1

$x = 7t$ $= 10$

A1F

(Accept 10.01 if t = 1.43 used)

***Alternative**: (use of symmetry)*

$v = 0$, $0 = 7 - 9.8t$

M1

$t = \frac{5}{7}$

A1

$x = 7 \times t \times 2$

m1

$x = 10$ (acc 9.996)

A1F

***Alternative**: (use of equation for Range)*

$R = \frac{2V^2 \sin\alpha\cos\alpha}{g}$

Correct formula used M1, 45° used m1

Correct subs A1,

result A1F, AWRT 10m, ft one slip.

(e) Speed^2 = $7^2 + 7^2$

M1

= 9.90 ms^{-1}

Accept 9.9, 7√2,
or 9.91 if t = 1.43 used in downward motion

A1

2

[16]

Q9.

(a) $\mathbf{v} = (2t - 6)\mathbf{i} + t^2\mathbf{j}$

differentiation

M1 A1, A1

3

(b) $2t - 6 - 0$

M1

$t = 3$

ft differentiation for t > 0

A1F

2

(c) $\mathbf{a} = 2\mathbf{i} + 2t\mathbf{j}$

M1

a is function of t, not constant

Differentiation of 2 components of vector expression for v attempted
*CAO **a** correct and explanation correct; reference to dependence on t required*

A1

2

[7]

Q10.

(a) Vertical: $0 = 4Ut - \frac{1}{2}gt^2$

Accept use of 9.8

M1A1

$t = \frac{8U}{g}$

A1

3

(b) horizontal: $OH = 3U \times \frac{8U}{g}$

Use of v × t

M1

$OH = \frac{24U^2}{g}$

A1

2

(c) speed$^2 = (3U)^2 + (4U)^2$

M1

speed $= 5U$

A1

2

(d) $\left(\sqrt{18}U\right)^2 = (3U)^2 + v^2$

M!A1

$v = (\pm)3U$

A1

$\pm 3U = 4U - gt$

Or attempt to solve quadratic in t

M1

$t_1 = \frac{U}{g}$ $\quad t_2 = \frac{7U}{g}$

A1A1

6

[13]

Q11.

(a) $-16i + 16j = \frac{1}{2}\mathbf{a} \times 4^2$

Use of vector constant acceleration equation

M1

Correct equation

A1

$\mathbf{a} = \frac{-16\mathbf{i} + 16\mathbf{j}}{8} = -2\mathbf{i} + 2\mathbf{j}$

Solving for ***a***

M1

ag Correct ***a***

A1

4

(b) $\mathbf{v} = 5(-2\mathbf{i} + 2\mathbf{j}) = -10\mathbf{i} + 10\mathbf{j}$

Expression for ***v*** *at t = 5*

M1

Correct expression

A1

$v = \sqrt{10^2 + 10^2} = 14.1 \text{ ms}^{-1}$ (to 3 sf)

Finding magnitude

M1

Correct speed

A1

4

(c) $20\mathbf{i} - 10\mathbf{j} + \mathbf{Q} = 3(-2\mathbf{i} + 2\mathbf{j})$

Application of Newton's second law to form equation

M1

Correct equation

A1

$\mathbf{Q} = (-20 - 6)\,\mathbf{i} + (10 + 6)\,\mathbf{j} = -26\mathbf{i} + 16\mathbf{j}$

Solving for ***Q***

M1

Correct Q

A1

4

[12]

Q12.

(a) $\mathbf{v} = -4e^{-t}\,\mathbf{i} + (6 - 3e^{-t})\mathbf{j}$

Differentiating position vector

M1

$t = 0$ A1

Correct velocity

$\mathbf{v} = -4\mathbf{i} + 3\mathbf{j}$

ag Substituting t = 0 to obtain initial velocity

A1

3

(b) $\mathbf{a} = 4e^{-t}\,\mathbf{i} + 3e^{-t}\,\mathbf{j}$

Differentiating velocity

M1

Correct acceleration

A1

2

(c) $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$

Finding acceleration when t = 0

M1

$a = \sqrt{4^2 + 3^2} = 5$

Correct magnitute

A1

2

(d) $\mathbf{v} \to 0\mathbf{i} + 6\mathbf{j}$

*For **i** component*

B1

*For **j** component*

B1

2

[9]

Q13.

(a) $\mathbf{v} = (3\mathbf{i} - 10\mathbf{j}) + (4\mathbf{i} + 2\mathbf{j})t$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1

$= (3 + 4t)\,\mathbf{i} + (2t - 10)\,\mathbf{j}$

ag *Correct result from correct working*

A1

2

(b) $2t - 10 = 0$

j *component equal to zero*

M1

$t = 5$

Correct time

A1

2

(c) (i) $\mathbf{r} = (3\mathbf{i} - 10\mathbf{j}) \times 10 + \frac{1}{2}(4\mathbf{i} + 2\mathbf{j}) \times 10^2$

Finding ***r*** *when t = 10*

M1

Correct expression

A1

$= 230\,\mathbf{i}$

Correct final answer

A1

3

(c) (ii) $\mathbf{v} = (3 + 4 \times 10)\,\mathbf{i} + (2 \times 10 - 10)\,\mathbf{j}$

$= 43\mathbf{i} + 10\mathbf{j}$

Correct velocity

B1

1

(d) $\mathbf{r} = 230\mathbf{i} + (43\mathbf{i} + 10\mathbf{j}) \times 10$

Uses zero acceleration

M1

Uses both answers from (c)

M1

$= 660\mathbf{i} + 100\mathbf{j}$

Correct expression for ***r***

A1

Correct simplified result

A1

r = $\sqrt{660^2 + 100^2}$

Finding magnitude

M1

= 668 m

Correct distance
Follow through from part (c)

A1 √

6

[14]

Q14.

(a) $5\mathbf{i} - 2\mathbf{j} = 4\mathbf{i} + 3\mathbf{j} + 10\mathbf{a}$

Use of constant acceleration equation for velocity

M1

Correct equation

A1

$a = \frac{1}{10}(\mathbf{i} - 5\mathbf{j}) = (0.1\mathbf{i} - 0.5\mathbf{j})\ \text{ms}^{-2}$

ag *Correct acceleration from correct working*

A1

3

(b) $\mathbf{r} = (4\mathbf{i} + 3\mathbf{j})t + 0.5(0.1\mathbf{i} - 0.5\mathbf{j})t^2$

Use of constant acceleration equation for position

M1

Correct expression

A1

2

(c) $\mathbf{r} = (4t + 0.05t^2)\mathbf{i} + (3t - 0.25t^2)\mathbf{j}$

j *component is zero*

M1

Correct equation

A1

$3t - 0.25t^2 = 0$

$t(3 - 0.25t) = 0$

Solving for t

m1

$t = 0$ or $t = \dfrac{3}{0.25} = 12$ s

$t = 12$ s

Correct time

A1

4

(d) $\mathbf{v} = (4 + 0.1t)\mathbf{i} + (3 - 0.5t)\mathbf{j}$

$3 - 0.5t = 0$

$t = 6$

Expression for velocity

M1

Correct velocity

A1

***j** component is zero*

m1

Correct time

A1

4

[13]

Q15.

(a) $60\mathbf{i} + 20\mathbf{j} = 20(2\mathbf{i}\ 3\mathbf{j}) + 200\mathbf{a}$

$20\mathbf{i} + 80\mathbf{j} = 200\mathbf{a}$

$\mathbf{a} = 0.1\mathbf{i} + 0.4\mathbf{j}$

*Use of constant acceleration equation in vector form to find **a***

M1

Correct equation

A1

*Correct **a***

A1

3

(b) $\mathbf{v} = (2\mathbf{i} - 3\mathbf{j}) + (0.1\mathbf{i} + 0.4\mathbf{j})t$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1

$= (2 + 0.1t)\mathbf{i} + (-3 + 0.4t)\mathbf{j}$

Correct expression

A1

2

(c) $2 + 0.1t = -(-3 + 0.4t)$

Equating components with ±

M1

$0.5t = 1$

Correct equation

A1

$t = 2$

A1

$\mathbf{v} = 2.2\mathbf{i} - 2.2\mathbf{j}$

Finding velocity

M1

$v = \sqrt{2.2^2 + 2.2^2} = 3.11 \text{ms}^{-1}$

Correct speed

A1

5

[10]

Q16.

(a) $19\mathbf{i} - 25\mathbf{j} = \frac{1}{2}\mathbf{a} \times 10^2 + 9\mathbf{i} + 10\mathbf{j}$

Using both position vectors to form a constant acceleration equation

M1

$50\mathbf{a} = 10\mathbf{i} - 35\mathbf{j}$

Correct equation

*Solving for **a***

A1 M1

a = 0.2**i** – 0.7**j**

Correct **a**

A1

4

(b) **v** = 10(0.2**i** – 0.7**j**)

use of **v** = **a***t*

M1

= 2**i** – 7**j**

Correct velocity

A1

$v = \sqrt{2^2 + 7^2}$ = 7.28 ms^{-1}

Finding speed from velocity

Correct speed

m1 A1

4

(c) 15.4 = $\frac{1}{2}$ × 0.2 × t^2 + 9

Finding t from one component

M1

t = 8

Correct t

A1

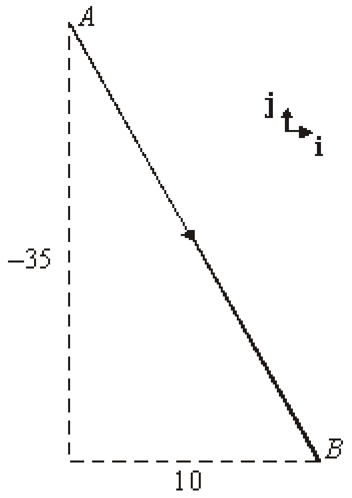
$\frac{1}{2}$ × (–0.7) × 8^2 + 10 = –12.4

Using t = 8 with other component

Correct result

M1 A

4

(d) 

Straight line
Correct direction

B1 B1

2

[14]

Q17.

(a) $40\mathbf{i} - 15\mathbf{j} = \frac{1}{2}(5\mathbf{i} - 2\mathbf{j} + \mathbf{v}) \times 10$

Use of constant acceleration equation with **v** *unknown*
Correct equation

M1 A1

$\mathbf{v} = 8\mathbf{i} - 3\mathbf{j} - 5\mathbf{i} + 2\mathbf{j} = 3\mathbf{i} - \mathbf{j}$

Solving for **v**
Correct **v**

M1 A1

4

(b) $3\mathbf{i} - \mathbf{j} = 5\mathbf{i} - 2\mathbf{j} + 10\mathbf{a}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1A1

$\mathbf{a} = -0.2\mathbf{i} + 0.1\mathbf{j}$

Correct **a**

A1

3

(c) $\mathbf{F} = 15(-0.2\mathbf{i} + 0.1\mathbf{j})$

Use of $\mathbf{F} = m\mathbf{a}$

M1

= –3**i** + 1.5**j**

Correct **F**

A1

$F = \sqrt{3^2 + 1.5^2}$ = 3.35

Finding magnitude of **F**

Correct magnitude

M1 A1

4

[11]

Q18.

(a) (i) **a** = $\frac{1}{4}$ (8**i** – 3**j**) = 2**i** – 3**j**

Use of ***F*** *= m**a**, must be applied to both components*

M1

Correct acceleration

A1

2

(ii) **v** = 20(2**i** – 3**j**) = 40**i** – 60**j**

Use of ***v*** *=* ***u*** *+* ***a****t with* ***u*** *= 0**i** + 0**j***

M1

Correct ***v***

A1

2

(iii) **r** = $\frac{1}{2}$ (2**i** – 3**j**) × 20^2

Use of constant acceleration equation to find ***r*** *with* ***u*** *= 0**i** + 0**j***

M1

Correct expression

A1

= 400**i** – 600**j**

Correct ***r*** *in simplified form*

A1

3

(b) **r** = 400**i** – 600**j** + 25(40**i** – 60**j**)

Use of ***r*** *+ 25****v***

M1

Correct expression

A1ft

= 1400**i** – 2100**j**

Correct position vector

A1ft

$r = \sqrt{1400^2 + 2100^2}$ = 2520 m (to 3 sf)

Finding magnitude

m1

Correct distance from correct working

A1ft

5

Alternative: *Straight line method*
M1 two distances r and 25v
A1 for each distance
m1 adding
A1 correct final answer

[12]

Q19.

(a) **r** = 5**j**

Correct vector

B1

1

(b) $4t - 0.01t^2 = 0$

Equation based on ***i*** *component*

M1

Correct equation

A1

$t = 0$ or $t = 400$

Solving the quadratic

M1

$t = 400$

Selecting t = 400

A1

4

(c) $\mathbf{v} = (4 - 0.02t)\mathbf{i} + (-3 - 0.08t)\mathbf{j}$

Use of $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{a}t$

M1

Correct velocity vector

A1

$4 - 0.02t = 3 + 0.08t$

Equation using both components

M1

Correct equation

A1

$t = 10$

Correct t

A1

5

[10]

Q20.

(a) $19\mathbf{i} + 13\mathbf{j} = 35\mathbf{i} + 45\mathbf{j} + 8\mathbf{a}$

Constant acceleration equation to find a

M1

$$\mathbf{a} = \frac{19-35}{8}\mathbf{i} + \frac{13-45}{8}\mathbf{j} = 2\mathbf{i} - 4\mathbf{j}$$

Correct equation

A1

ag *Correct* ***a*** *from correct working*

A1

3

(b) $\mathbf{r} = (35\mathbf{i} + 45\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} - 4\mathbf{j})t^2$

Use of constant acceleration equation

M1

Correct ***i*** *component*

A1

Correct ***j*** *component*

A1

3

(c) $\mathbf{r} = (3.5t - t^2)\mathbf{i} + (45t - 2t^2)\mathbf{j}$

Splitting into components correctly

B1

$35t - t^2 = 300$

Forming equation for one component

M1

$t^2 - 35t + 300 = 0$

Solving quadratic

M1

$t = 15$ or $t = 20$

Two correct solutions

A1

$4t - 2t^2 = 225$

Forming correct second quadratic

B1

$2t^2 - 45t + 225 = 0$

$t = 7.5$ or t = 15

Solving quadratic for two solutions

M1

$t = 15$ seconds

Correct final solution

A1

7

[13]

Q21.

(a) $\mathbf{F} = (3\mathbf{i} + 4\mathbf{j}) + (6\mathbf{i} - 8\mathbf{j})$

Addition of the two forces

M1

2

$= 9\mathbf{i} - 4\mathbf{j}$

Correct resultant

A1

(b) $F = \sqrt{9^2 + 4^2} = 9.85\ \text{N}$

Finding magnitude

M1

Correct magnitude

A1

2

(c) $\tan\alpha = \frac{4}{9}$

Using tan to find the angle

M1

Correct equation

A1

$\alpha = 24.0°$

Correct angle

A1

3

[7]

Q22.

(a) 2**i** + 14**j** = **u** + 2(2**i** – 3**j**)

*Using constant acceleration equation to find **u***
Correct equation

M1

*Correct **u** from correct working*

A1

u = (2 – 4) **i** + (14 + 6)**j** = – 2**i** + 20**j**)

A1

3

(b) **r** = (– 2**i** +20**j**) × 3 + $\frac{1}{2}$ (2**i** – 3**j**) × 9

*Using constant acceleration equation to find **r***
Correct equation

M1

*Correct **r***

A1

$3\mathbf{i} + 46.5\mathbf{j}$

A1

$r = \sqrt{3^2 + 46.5^2} = 46.6\,\text{ms}^{-1}$

Finding magnitude
Correct magnitude

M1

5

(c) $\mathbf{v} = (2T - 2)\mathbf{i} + (20 - 3T)\mathbf{j}$

*Forming equation for **v***

M1

$100 = (2T - 2)^2 + (20 - 3T)^2$

Forming equation that T must satisfy

M1

$13T^2 - 128T + 304 = 0$

Correct Equation

A1

$$T = \frac{128 \pm \sqrt{128^2 - 4 \times 13 \times 304}}{2 \times 13}$$

Solving for T

M1

= 4 or 5.85

Correct solutions

A1

$T = 4$

Selecting T = 4

A1

6

[14]

Q23.

(a) $\mathbf{a} = \dfrac{1000t\mathbf{i} - 5000\mathbf{j}}{2000} = \dfrac{t}{2}\mathbf{i} - \dfrac{5}{2}\mathbf{j}$

Using Newton's 2nd law

M1

Correct acceleration

A1

2

(b) $\mathbf{v} = \left(\frac{t^2}{4} + c\right)\mathbf{i} + \left(\frac{-5t}{2} + d\right)\mathbf{j}$

Integration

M1

Correct integral with or without constants

A1

$t = 0$, $\mathbf{v} = 6\mathbf{j} \Rightarrow c = 0, d = 6$

Evaluation of constants

m1

Correct constants and correct final answer

A1

4

$\mathbf{v} = \left(\frac{t^2}{4}\right)\mathbf{i} + \left(6 - \frac{5t}{2}\right)\mathbf{j}$

(c) $\mathbf{r} = \left(\frac{t^3}{12} + e\right)\mathbf{i} + \left(6t - \frac{5t^2}{4} + f\right)\mathbf{j}$

Integration

M1

Correct integral with or without constants

A1

$t = 0$, $\mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow e = 0, f = 0$

Evaluation of constants

m1

$\mathbf{r} = \left(\frac{t^3}{12}\right)\mathbf{i} + \left(6t - \frac{5t^2}{4}\right)\mathbf{j}$

Correct constants and correct final answer

A1

4

[10]

Q24.

(a) $\mathbf{r} = (4\mathbf{i} + 7\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} + 4\mathbf{j})t^2$

*Finding **r***

M1

$= (4t = t^2)\mathbf{i} + (7t - 2t^2)\mathbf{j}$

*Correct **r***

A1

2

(b) $4t - t^2 = 3$

Forming equation for one component

M1

$t^2 - 4t + 3 = 0$

$(t - 1)(t - 3)$

Solving quadratic equation

m1

$(t - 1)(t - 3)$

Solving quadratic equation

m1

$t = 1$ or $t = 3$

Two solutions

A1

$7t - 2t^2 = 5$

Forming and solving quadratic equation

M1

$2t^2 - 7t + 5 = 0$

$(2t - 5)(t - 1)$

$t = 1$ or $t = 2.5$

Two solutions

A1

$t = 1$

Selecting correct solution

A1

6

[8]

Q25.

(a) (i) $\vec{AB} = (70\mathbf{i} - 40\mathbf{j}) - (30\mathbf{i} - 35\mathbf{j}) = 40\mathbf{i} - 5\mathbf{j}$

cao

B1

1

(ii) $40\mathbf{i} - 5\mathbf{j} = 10\mathbf{v} + \frac{1}{2}(0.2\mathbf{i} + 0.1\mathbf{j}) \times 100$

Using their $\vec{AB}$

Correct RHS of the equation

M1 A1

$10\mathbf{v} = 30\mathbf{i} - 10\mathbf{j}$

correct $\boldsymbol{v}$

$\boldsymbol{v} = 3\mathbf{i} - \mathbf{j}$

A1

3

(b) $3\mathbf{i} - \mathbf{j} = \mathbf{u} + (0.2\mathbf{i} + 0.1\mathbf{j}) \times 10$

Correct RHS

M1 A1

$\mathbf{u} = \mathbf{i} - 2\mathbf{j}$

Correct $\boldsymbol{u}$

A1

3

(c) $30\mathbf{i} - 35\mathbf{j} = (\mathbf{i} - 2\mathbf{j}) \times 10 + \frac{1}{2}(0.2\mathbf{i} + 0.1\mathbf{j}) \times 100 + \mathbf{r}0$

Equation of correct form for t = 10 or 20 with their $\boldsymbol{u}$

Correct equation with their $\boldsymbol{u}$

M1 A1

$30\mathbf{i} - 35\mathbf{j} = 20\mathbf{j} - 15\mathbf{j} + \mathbf{r}_0$

$\mathbf{r}_0 = 10\mathbf{i} - 20\mathbf{j}$

Correct position vector

A1

3

[10]

Q26.

(a) $\mathbf{v} = -2\sin\left(\frac{t}{20}\right)\mathbf{i} + 4\cos\left(\frac{t}{20}\right)\mathbf{j}$

Differentiating

i *component correct*

j *component correct*

M1 A1 A1

3

(b) $\mathbf{v} = 4\mathbf{j}$

Correct velocity at t *= 0*

B1

Travelling north

Travelling north with **v** = n***j*** *where* n > *0*

B1

2

(c) $2\sin\left(\frac{t}{20}\right) = 0$

Equation for t *based on 0**i***

M1

$\frac{t}{20} = \pi$

$t = 20\pi$

Correct solution

A1

2

[7]

Q27.

(a) $\mathbf{F} = (2 + 6 - 2)\mathbf{i} + (3 - 8 + 10)\mathbf{j}$

Summing the forces

M1

$= 6\mathbf{i} + 5\mathbf{j}$

Correct resultant

A1 **2**

(b) $\mathbf{a} = \frac{1}{8}(6\mathbf{i} + 5\mathbf{j}) = 0.75\mathbf{i} + 0.625\mathbf{j}$

Dividing by 8 to get correct value

B1 **1**

(c) (i) $\mathbf{v} = (3\mathbf{i} - 2\mathbf{j}) + 40(0.75\mathbf{i} + 0.625\mathbf{j})$

use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1

$= 33\mathbf{i} + 23\mathbf{j}$

Correct $\mathbf{v}$

A1 **2**

(ii) $\mathbf{r} = (3\mathbf{i} - 2\mathbf{j}) \times 40 + \frac{1}{2}(0.75\mathbf{i} + 0.625\mathbf{j}) \times 40^2$

Vertex constant acceleration equation to find $\mathbf{r}$

Correct equation

M1 A1

$= 720\mathbf{i} + 420\mathbf{j}$

Correct position vector

A1

$r = \sqrt{720^2 + 420^2} = 834$ m

Finding magnitude

Correct magnitude

M1 A1 **5**

[10]

Q28.

(a) $44\mathbf{i} + 28\mathbf{j} = 10(4\mathbf{i} + 2\mathbf{j}) + 50\mathbf{a}$

Equation to find ***a*** *using position vector*

Correct equation

M1/A1

a = 0.08**i** + 0.16**j**

*Solving for **a***

*Correct **a***

M1/A1

4

(b) **r** = (4**i** + 2**j**)t + $\frac{1}{2}$ (0.08**i** + 0.16**j**)t^2

*Equation for **r***

One component correct

M1/A1

= (4t + 0.04t^2)**i** + (2t + 0.08t^2)**j**

Second component correct

A1

3

(c) (i) $96 = 4T + 0.04T^2$ $\quad 72 = 2T + 0.08T^2$

Forms and solves equation for one component

M1

T = 20 or –120 $\quad T$ = 20 or –45

Correct solutions

A1

Hence T = 20

Forms and solves equation for other component

Correct solutions plus correct conclusion

M1/A1

4

(d) **v** = (4**i** + 2**j**) + (0.08**i** + 0.16**j**) × 20

*Finding **v** at* t = *20*

M1

= 5.6**i** + 5.2**j**

Correct v

A1

$v = \sqrt{5.6^2 + 5.2^2} = 7.64\ ms^{-1}$

Finding correct speed

A1

3

[14]